

Mission 2 — Mechanical Design Core Skills

Step 1: Plan Submission

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1. What I Will Build

I am going to design a **mechanical mounting bracket** in Autodesk Fusion with exact dimensions, mounting holes, and filleted edges. This is a real-world mechanical component used to mount and secure parts in machines, panels, and assemblies.

Component	Specification
Shape	Rectangular Bracket
Dimensions	100mm x 60mm x 8mm
Mounting Holes	4 corner holes — 8mm diameter
Center Hole	1 clearance hole — 20mm diameter
Edge Finish	2mm Fillet on all edges
Material	Aluminum (visual)

2. How I Will Do It — Step-by-Step

- Step 1:** Open Autodesk Fusion and create a new design file
- Step 2:** Click Sketch and select the XY plane as the drawing surface
- Step 3:** Draw a rectangle of 100mm x 60mm using the sketch tools
- Step 4:** Add 4 circles (8mm diameter) at each corner, 10mm from edges
- Step 5:** Add 1 center circle (20mm diameter) in the middle of the bracket
- Step 6:** Apply dimensions and constraints to fully define the sketch
- Step 7:** Finish Sketch and Extrude the profile to 8mm height
- Step 8:** Use Extrude with Cut operation to punch holes through the bracket
- Step 9:** Apply 2mm Fillet on all sharp edges for a clean finish
- Step 10:** Save to cloud, take screenshots, and document the work

3. Tools I Will Use

Tool	Purpose	Cost
Autodesk Fusion	Primary CAD tool for modeling	Free (personal)
GitHub	Documentation and submission	Free
Google Drive	Backup screenshots and files	Free

4. Time Estimate

Phase	Estimated Time
Sketching & Dimensions	45 minutes
Extrusion & Hole Cutting	30 minutes
Fillets & Appearance	20 minutes
Screenshots & Documentation	30 minutes
GitHub Upload & Submission	15 minutes
Total	~3 hours